

EdgeResilience

Snapdragon-Powered Predictive Intelligence for Connected Vehicle Safety

Repository: github.com/Sureshdmesw/EdgeResilience | Evidence: Snapdragon X Elite CRD (SC8380XP), Qualcomm AI Hub, QNN HTP

1. Executive Summary

Connected vehicles increasingly depend on cloud and infrastructure services, yet the intelligence that protects them must keep working when those links degrade or fail. **EdgeResilience** is a locally executed, predictive resilience pipeline. A compact Temporal Transformer forecasts near-future degradation of vehicle cyber-physical health from a 12-step window of 17 telemetry features. A deterministic policy converts the forecast into a risk level, and a connectivity-aware runtime keeps inference active on the device, preserves evidence while connectivity is impaired, and prepares an orderly hand-over when connectivity returns.

The V4 model was compiled through Qualcomm AI Hub and executed on a **Snapdragon X Elite CRD (SC8380XP)** using the Qualcomm QNN HTP (NPU) backend. All **90 of 90** profiled execution-detail entries ran on the NPU, with an estimated inference latency of **63 µs** and a peak inference memory of **28.5 MB**. The prototype is a software simulation trained on a synthetic dataset. The full CPU-to-Snapdragon numerical comparison did not meet its acceptance threshold; this is reported openly in Sections 7 and 9 and is the first item of future work.

2. Problem Statement

Connected vehicles face cyber threats such as CAN bus anomalies, message injection, timing manipulation and payload tampering. Three operational needs follow. The system should **anticipate** degradation before it becomes critical, **remain intelligent** using only local compute when a cloud or infrastructure backend is unreachable, and **preserve auditable evidence** of anomalous behaviour for post-event analysis. Analytics that depend on a live backend cannot be assumed available at the moment they are needed most.

3. Proposed Solution and Differentiation

EdgeResilience is an end-to-end edge pipeline: telemetry window, temporal prediction, deterministic risk interpretation, connectivity evaluation, local inference, evidence buffering, recovery detection and a synchronisation manifest. Its distinguishing design choices are:

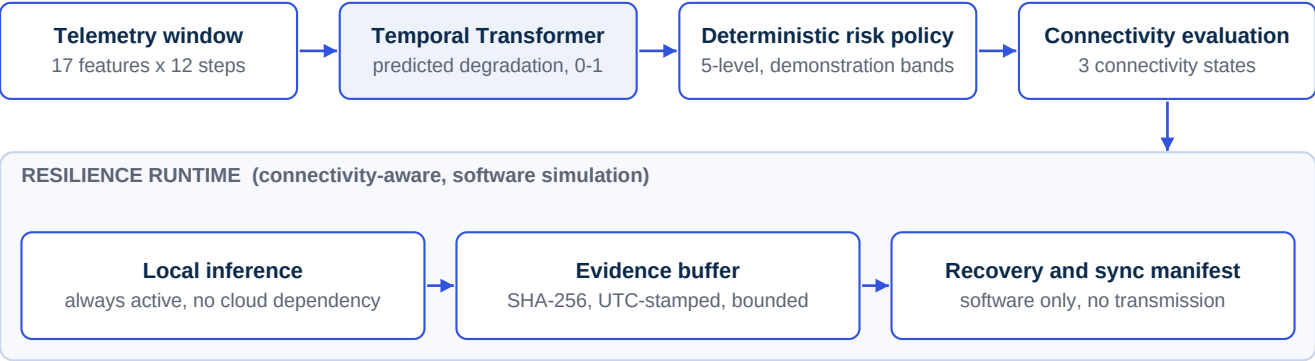
- Predictive rather than purely reactive.** The model estimates future degradation from temporal context, not only current state.
- Connectivity-independent by design.** Local inference remains active in every connectivity state.
- Separation of AI and policy.** The probabilistic model produces a bounded score; every risk, buffering and synchronisation decision is made by an inspectable deterministic layer. No AI output commands vehicle actuation.
- Evidence continuity.** SHA-256-hashed, UTC-stamped records are buffered during degraded or disconnected operation and listed in a synchronisation manifest after recovery.
- Evidence discipline.** Each reported result is tied to a reproducible artefact in the public repository, and unresolved items are stated as such.

4. AI / ML Architecture

Property	Specification
Model	EdgeResilienceTemporalPredictor (V4), Temporal Transformer
Input / output	12 steps x 17 features, float32; one predicted future-degradation score in [0, 1]
Structure	Input projection to d_model 64, learned positional embedding, 2 encoder layers, 4 attention heads, feed-forward width 128, learned temporal attention pooling, compact output head with sigmoid
Size	71,170 parameters
Accuracy (held-out test split)	MAE approximately 0.0051; RMSE approximately 0.0066
Training data	5,000 synthetic temporal scenarios (dataset SHA-256 recorded in the repository)
Ablation	The full 12-step sequence outperformed both final-step-only and compact-summary inputs

Accuracy figures describe performance on synthetic scenario data and should not be read as real-vehicle detection performance.

5. EdgeResilience System Architecture



The risk policy (*V4_DEGRADATION_DEMO_POLICY_V1*) maps the score to NORMAL, LOW, MEDIUM, HIGH or CRITICAL using fixed demonstration thresholds; these are not certified safety limits. Connectivity is classified from heartbeat, packet-loss and latency conditions. The evidence buffer is bounded and canonicalised. The synchronisation manifest lists event identifiers, payload digests and sequence numbers; no network transmission takes place. A three-cycle demonstration (`python demo/run_demo.py`) exercises the complete path: connected operation, a disconnection during which evidence is preserved, and a recovery in which the manifest becomes ready.

6. Snapdragon Optimization and Deployment

The deployment path is PyTorch checkpoint, dynamic-batch ONNX export, QNN-ready ONNX derivative, Qualcomm AI Hub compilation, and profiled execution on the Snapdragon X Elite CRD through QNN on the Hexagon HTP. The ONNX export matches the PyTorch reference within a maximum absolute error of 5.96e-08 against a 1e-5 threshold. For the QNN-ready derivative, constant folding removed the Mod operators and reduced the graph from 336 to 176 nodes, leaving only standard ONNX opset 17 operators; its outputs are identical to the source ONNX model on CPU across 200 samples.

With 71,170 parameters and a 28.5 MB peak inference footprint, the model suits always-on, low-latency inference on the NPU of a Snapdragon-class device. **Provenance.** Model development and CPU reference measurements were performed on a Lenovo IdeaPad 3 15IAU7 (Intel Core i5-1235U). Snapdragon compilation, execution and profiling used Qualcomm AI Hub with a Snapdragon X Elite CRD. The design targets Snapdragon-powered HP PCs; all current hardware evidence is from the CRD, and no measurement on an HP Snapdragon device is claimed.

7. Validated Performance

Metric (Snapdragon X Elite CRD, SC8380XP, QNN HTP)	Result
Profiled execution-detail entries on NPU	90 of 90
Estimated inference latency	63 μs
Warm latency: median / mean	67 μs / 68.96 μs
Warm latency: P90 / P95 / P99	77 / 82 / 88 μs
Peak inference memory	28.5 MB
Warm profile-derived rate (single batch)	approximately 14,501 inferences per second

These are workload-specific profiling results for the model alone. They are not end-to-end vehicle-system latency. For context, the ONNX Runtime CPU baseline on the development laptop measured 0.267 ms mean at batch size 1; it was obtained with a different method and is not presented as a Snapdragon speed-up.

Numerical fidelity. The complete CPU-to-Snapdragon comparison was executed but did not meet the acceptance threshold (maximum absolute error 0.0030923495; maximum relative error 38.50%). EdgeResilience therefore makes no claim of end-to-end numerical equivalence between the CPU reference and the Snapdragon execution. Because the model output lies in [0, 1] and the demonstration risk bands are only 0.01 to 0.02 wide, an error of this size could move a score across a band boundary, which is why closing this gap is the first development priority.

8. Safety, Cybersecurity and Responsible Deployment

- Software simulation only.** No physical vehicle, production vehicle connection, external CAN transmission, direct actuation, steering, braking or throttle control, or emergency dispatch is involved.

- **Deterministic safety-relevant logic.** The AI layer outputs a score; risk level, evidence requirement and synchronisation eligibility are rule-based and auditable.
- **Local-first evidence handling.** Records are hashed and timestamped on the device, and synchronisation is prepared only after recovery is detected.
- **No certification claim.** Risk thresholds are demonstration defaults. Conformance to ISO 26262 or ISO/SAE 21434 is not claimed and would require a formal safety and cybersecurity case before any production consideration.

9. Evidence and Current Validation Boundaries

Validated and reported	Not claimed
PyTorch reference and ONNX export equivalence; QNN-ready graph derivative	End-to-end CPU-to-Snapdragon numerical equivalence
Qualcomm AI Hub compilation; QNN HTP execution on Snapdragon X Elite CRD (SC8380XP); 90 of 90 profiled entries on NPU	HTP utilisation percentage
Latency, memory and rate figures in Section 7	Project-specific power or thermal measurements
Three-cycle software demonstration of prediction, local inference, evidence buffering, recovery and manifest preparation	Validation on a physical HP Snapdragon laptop, a physical vehicle, or real vehicle telemetry
Reproducible artefacts and provenance register in the public repository	Direct vehicle actuation or deployment as a production automotive safety controller

10. Future Development

- **Numerical closure.** Investigate operator-level divergence, apply corrective graph or precision changes, re-run the full comparison against the 1e-5 threshold and refresh the Snapdragon profile.
- **Deployment characterisation.** Measure on a Snapdragon-powered HP PC, including HTP utilisation, power and thermal behaviour where tooling allows.
- **Data realism.** Evaluate on real or public automotive telemetry and calibrate the risk policy with domain experts.
- **System integration.** Progress to controlled hardware-in-the-loop evaluation and, later, multi-vehicle V2X cooperative resilience in simulation.

11. Conclusion

EdgeResilience demonstrates a complete predictive, connectivity-independent resilience pipeline whose core model executes on the Snapdragon X Elite NPU with microsecond-scale estimated latency and a small memory footprint. Its value to automotive and platform teams lies in a clear architecture, a strict separation between AI and safety-relevant logic, and evidence that is reproducible and honestly bounded. We would welcome the opportunity to present the work and to close the remaining validation items with Snapdragon platform expertise.